# Design Document (DD)

## An Energy-Efficient Dual Hydroponic System for Organic/Inorganic Capsicum Chili Farming

## Project ID: 25-26J-035

**Individual component: Design and Implement a Semi -Automated Inorganic NFT Hydroponic System**

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# 1. Introduction

## 1.1 Purpose

The purpose of this Document is to present the detailed design of the Semi-Automated inorganic Nutrient Film Technique (NFT) hydroponic subsystem developed as part of An Energy-Efficient Dual Hydroponic System for Organic/Inorganic Capsicum Chili Farming final-year research project. This document focuses on the architectural design, interfaces, control logic, and system constraints of the embedded IoT-based nutrient management solution. The document serves as a technical reference for implementation, evaluation, and future enhancement of the system.

## 1.2 Scope

This Design Document covers the design and implementation of the Semi-Automated inorganic NFT hydroponic subsystem. Specifically, the scope of this document includes:

- Inorganic NFT hydroponic structure implementation
- Sensor integration and real-time data acquisition
- ESP32 firmware development for reading and processing sensor data
- Cloud-based data upload using Firebase Realtime Database
- Mobile and dashboard-based visualization of sensor readings and system status

The scope of this document does not currently include the following components:

- Filtration and sterilization mechanisms for the nutrient solution
- fuzzy logic based nutrient dosing decisions

These excluded components are planned for subsequent project milestones and future system enhancements.

## 1.3 Definitions, Acronyms, and Abbreviations

| Term | Full Meaning | Definition and Context |
|---|---|---|
| **NFT** | Nutrient Film Technique | A hydroponic system where a thin film of nutrient solution recirculates over plant roots. |
| **EC** | Electrical Conductivity | A measure of the ionic strength of a solution, used to estimate nutrient concentration. |
| **pH** | Potential of Hydrogen | A scale (0-14) representing the acidity or alkalinity of the nutrient solution. |
| **NPK** | Nitrogen, Phosphorus, Potassium | The primary macronutrients required for plant growth and development. |
| **ESP32** | Espressif Systems 32 | A low-power, dual-core microcontroller with integrated Wi-Fi and Bluetooth. |
| **RS485** | Recommended Standard 485 | A standard for serial communication transmission lines in industrial IoT. |
| **Modbus RTU** | Remote Terminal Unit | A robust communication protocol used to poll sensors via the RS485 interface. |
| **Firebase** | Google Firebase | A platform for real-time cloud database storage and dashboard hosting. |

## 1.4 Overview

The Semi-Automated Inorganic NFT Hydroponic Nutrient Management System is an embedded IoT-based solution designed to monitor and control nutrient conditions for Capsicum chili cultivation using the Nutrient Film Technique. The system continuously measures key parameters such as electrical conductivity (EC), pH, water level, and plant height, and automatically regulates nutrient strength and acidity through controlled dosing mechanisms executed by an ESP32 microcontroller. Its main objective is to maintain a stable and safe environment while minimizing manual intervention and operational errors. The system performs real-time sensing, decision-making, actuation, and data logging, with system status and historical data made available through a local LCD and a cloud-based dashboard. The solution is intended for small-scale hydroponic operators and research-based agricultural environments where reliability, simplicity, and cost-effective automation are essential.

# 2. Overall Descriptions

The Semi-Automated Inorganic NFT Hydroponic Nutrient Management System is designed as a modular embedded control platform that integrates sensing, decision-making, actuation, and remote monitoring to support stable and efficient hydroponic cultivation. The system operates as a closed-loop solution where environmental and nutrient-related parameters are continuously observed and used to guide automated control actions, ensuring consistent growing conditions over extended periods of operation.

The system is structured around a central microcontroller that coordinates data acquisition from multiple sensors, processes the collected information using predefined control logic, and activates corrective mechanisms when deviations from acceptable operating ranges are detected. Emphasis is placed on reliability, safety, and predictability rather than aggressive optimization, making the system suitable for real-world deployment in small-scale and research-oriented hydroponic environments. The design prioritizes gradual adjustments, verification through re-measurement, and protective constraints to avoid instability or damage to plants and equipment.

In addition to local autonomous operation, the system supports remote visibility through cloud-based data synchronization, allowing users to observe system behavior, review historical trends, and respond to abnormal conditions without direct physical access. The overall architecture is intentionally modular and scalable, enabling future extensions such as advanced control algorithms, additional sensors, or integration with larger hydroponic installations without requiring fundamental redesign of the core system.

## 2.1 Product perspective

The Semi-Automated Inorganic NFT Hydroponic Nutrient Management System is positioned as an enhanced alternative to conventional manual NFT hydroponic systems commonly used in small-scale farming and research environments. Traditional NFT setups rely heavily on manual measurement and adjustment of nutrient strength and pH, which increases the risk of human error, delayed response to parameter deviations, and inconsistent plant growth. Some commercially available hydroponic controllers provide basic automation; however, they are often costly, limited in flexibility, and designed as closed systems that do not support customization or detailed monitoring at the subsystem level.

In contrast, the proposed system adopts an open, embedded-system-based architecture that integrates real-time sensing, autonomous control, and cloud-based monitoring within a single, low-cost platform. Unlike threshold-only monitoring solutions that focus primarily on data visualization, this system actively performs controlled corrective actions while enforcing safety constraints such as water-level protection and time-buffered dosing. The modular design allows the system to function independently as a complete nutrient management unit while remaining compatible with larger hydroponic installations and future functional extensions. This positioning makes the system particularly suitable for experimental, educational, and resource-constrained environments where reliability, transparency, and adaptability are prioritized over proprietary features.

### 2.1.1 System interfaces

- The system operates on an embedded microcontroller platform and does not require a conventional operating system.
- Control firmware runs directly on the ESP32 without an OS kernel.
- Firmware is developed and uploaded using the Arduino Integrated Development Environment (IDE).
- External access to system data is platform-independent and can be performed from any operating system via standard internet connectivity.

### 2.1.2 User interfaces

- The system provides both local and remote user interfaces for monitoring and basic control.
- A 16×2 I2C LCD is used as the local interface to display real-time inorganic system parameters such as EC, pH, NPK values, water level status, pump states, and system alerts.
- A mobile application dashboard provides a graphical remote interface to view live sensor readings, differentiate between organic and inorganic systems, and perform basic control actions.
- Only the interface concept and perspective are presented here; detailed user interface behaviour is described in Section 3.

### 2.1.3 Hardware interfaces

- **ESP32 microcontroller** used as the central processing and control unit.

- **RS485-based multi-parameter nutrient sensor** for EC, pH, and NPK measurements.
- **Ultrasonic sensor** for non-contact plant height measurement.
- **Water-level sensor** for safety monitoring and pump protection.
- **Relay modules** used to interface the microcontroller with high-voltage pumps.
- **Nutrient and pH dosing pumps** used for automated correction of nutrient solution parameters.
- **Nutrient solution water tanks** are used for storage and circulation of the inorganic nutrient solution.
- **PVC pipes and NFT channels** are used for nutrient flow and plant support.
- **Electrical wiring and connectors** used to interconnect sensors, actuators, and control units.

### 2.1.4 Software interfaces

- Embedded firmware developed using the **Arduino IDE** for sensor reading, control logic, and actuation.
- **Firebase Realtime Database** is used as the cloud-based software platform for storing sensor data, pump states, and system alerts.
- **Mobile application dashboard** interfacing with Firebase to display real-time and historical system data and to issue basic control commands.
- The system remains platform-independent, as Firebase supports data access across different operating systems.

### 2.1.5 Communication interfaces

- **Wi-Fi (IEEE 802.11)** used for external communication between the ESP32 and the cloud platform.
- **HTTP/REST protocols** used to transmit sensor data to and receive control commands from Firebase.
- **RS485 (Modbus protocol)** used for internal communication between the ESP32 and the multi-parameter nutrient sensor.

### 2.1.6 Memory constraints

- The embedded software operates within the **limited flash and RAM memory** available on the ESP32 microcontroller.
- Lightweight libraries and efficient data structures are used to minimize runtime memory usage.
- Sensor data is processed in real time, and **continuous buffering is avoided** to reduce RAM consumption.
- **No long-term data storage is maintained locally** on the ESP32.
- **Firebase Realtime Database is used as external memory** for storing historical sensor data, system logs, and status information, reducing internal memory requirements.

### 2.1.7 Operations

**Normal Operations**

- System power-on and sensor warm-up.
- Continuous sensing and real-time data acquisition of nutrient and environmental parameters.
- Automated evaluation of sensor readings and activation of nutrient and pH dosing pumps when required.
- Upload of sensor readings and system status to the cloud database.
- Real-time visualization of system data on the mobile/dashboard interface.

**Special Operations**

- Sensor calibration mode for maintenance and accuracy verification.
- Pump testing and maintenance mode to verify correct actuator operation.
- Automatic safety shutdown of pumps during low water level conditions.
- Safe monitoring mode when a sensor provides invalid or abnormal readings.
- Manual override of pump operation through the dashboard during maintenance or emergency situations.

### 2.1.8 Site adaptation requirements

- The system is intended for deployment in **small-scale indoor or semi-controlled hydroponic environments** using inorganic NFT techniques.
- Installation requires a **stable electrical power supply** for the controller, sensors, and pumps.
- **Wi-Fi connectivity** is required for cloud data synchronization and remote monitoring.
- Sensors must be **properly positioned** within the nutrient reservoir and NFT channels to ensure accurate measurements.
- System threshold values can be **configured through firmware settings** to adapt to different plant growth stages or site conditions without hardware modification.

## 2.2 Product functions

The Semi-Automated Inorganic NFT Hydroponic Nutrient Management System provides a set of core functions that collectively support stable and safe hydroponic cultivation. The primary function of the system is real-time monitoring of nutrient solution parameters, including electrical conductivity (EC), pH, nutrient composition (NPK), water level, and temperature. Based on these measurements, the system performs automated nutrient strength adjustment and pH correction using controlled dosing mechanisms.

In addition to nutrient regulation, the system enforces safety functions such as pump shutdown during low water conditions to prevent hardware damage and incorrect dosing. The system also supports plant growth observation through non-contact ultrasonic height measurement and provides continuous visibility of system status via a local display and a cloud-connected dashboard. All sensor readings, pump states, and alerts are logged to the cloud to enable remote supervision and historical analysis.

## 2.3 User characteristics

- **Undergraduate researchers** with basic knowledge of hydroponic systems and sensor-based monitoring.
- **Small-scale hydroponic operators** with practical experience in nutrient management and system maintenance.

- **Novice users** with limited technical background, who rely on automated operation after initial setup.
- **Technically proficient users** capable of performing sensor calibration, system configuration, and basic troubleshooting.

## 2.4 Constraints

- The system is constrained by the **limited processing power and memory resources** of the ESP32 microcontroller.
- Sensor accuracy and response time are limited by the **specifications of commercially available EC, pH, NPK, and ultrasonic sensors**.
- Reliable operation depends on **stable Wi-Fi connectivity** for cloud data synchronization and remote monitoring.
- The system uses **pre-mixed inorganic nutrient solutions (A+B)**, limiting fine-grained individual nutrient control.
- The implementation is **prototype-scale**, which restricts mechanical robustness and large-scale deployment.

## 2.5 Assumptions and dependencies

# 2.5 Assumptions and Dependencies

- The system assumes **continuous availability of electrical power** during normal operation.
- A **stable wireless (Wi-Fi) network connection** is assumed for cloud data synchronization and remote monitoring.
- All sensors are assumed to be **properly installed and calibrated** before system operation.
- The system assumes the use of **commercial inorganic A+B nutrient solutions** according to manufacturer guidelines.
- The system depends on the **availability and reliability of the Firebase cloud platform** for data storage and dashboard visualization.
- Future versions of the system are assumed to operate on **similar embedded IoT platforms with wireless connectivity**.

## 2.6 Apportioning of requirements

- **Phase 1 – Core System Functions**
    - Sensor data acquisition and real-time monitoring.
    - Safety enforcement mechanisms such as water-level protection.
    - Automated EC and pH control using dosing pumps.

- **Phase 2 – Monitoring and Connectivity**
    - Cloud-based data logging using Firebase Realtime Database.
    - Remote monitoring and basic manual control through the mobile dashboard.

- **Phase 3 – Future Enhancements**
    - Advanced analytics and intelligent control techniques.
    - Extended automation and system optimization features.

# 3. Specific requirements (DD)

## 3.1 External interface requirements

### 3.1.1 User interfaces

The system provides both local and remote user interfaces to support monitoring, supervision, and manual intervention.

**Local User Interface**

A 16×2 I2C Liquid Crystal Display (LCD) is directly connected to the ESP32 microcontroller and serves as the primary local interface. The LCD displays real-time information related to the inorganic hydroponic system, including:

- Electrical Conductivity (EC) value
- pH value
- NPK readings (Nitrogen, Phosphorus, Potassium)
- Water level status
- Plant height measurement
- Pump operational states
- System alerts and warning messages

The local interface is designed for quick, on-site observation and does not require network connectivity.

**Remote User Interface (Mobile Dashboard)**

A mobile application dashboard developed by the project team provides a graphical remote user interface. Within the dashboard, the inorganic system is clearly separated from the organic system and displays real-time sensor readings retrieved from the Firebase Realtime Database. Users can:

- View live and historical sensor data
- Identify system status through visual indicators
- Manually toggle nutrient and pH dosing pumps when required

- Monitor alerts and abnormal conditions

The remote interface enables supervision without physical access to the hydroponic system and is accessible through standard mobile devices.

### 3.1.2 Hardware interfaces

The embedded firmware interfaces with multiple hardware components essential for system operation. The ESP32 microcontroller acts as the central controller and communicates with the following hardware interfaces:

- **RS485 Multi-Parameter Sensor**: Used for EC, pH, NPK, temperature, and moisture measurements via Modbus communication.
- **Ultrasonic Sensor**: Interfaces through GPIO pins to measure plant height using distance calculation.
- **Water Level Sensor**: Provides digital or analog input to detect low nutrient solution levels.
- **Relay Modules**: Controlled via GPIO outputs to safely switch high-voltage nutrient and pH dosing pumps.
- **Nutrient and pH Dosing Pumps**: Actuated through relay isolation to prevent electrical interference with the microcontroller.

All hardware interfaces are designed to ensure electrical safety, signal stability, and reliable long-term operation in a hydroponic environment.

### 3.1.3 Software interfaces

The embedded control software interfaces with several external software systems to support data storage, visualization, and control:

- **Firebase Realtime Database**: Acts as the primary cloud-based data repository for sensor readings, pump states, alerts, and timestamps.
- **Mobile Application (Dashboard)**: Retrieves data from Firebase and issues control commands for manual pump operation.
- **Firmware Development Environment**: Arduino IDE is used to develop, compile, and upload firmware to the ESP32 microcontroller.

These software interfaces allow the system to remain platform-independent and enable seamless integration with web and mobile technologies.

### 3.1.4 Communication interfaces

The system uses both **local** and **network-based** communication interfaces:

- **Wi-Fi (IEEE 802.11)**: Enables internet connectivity for cloud synchronization between the ESP32 and Firebase.
- **HTTP/REST Protocols**: Used to transmit sensor data and receive control commands from the cloud platform.
- **RS485 (Modbus Protocol)**: Provides robust, noise-resistant communication between the ESP32 and the industrial multi-parameter nutrient sensor.

This hybrid communication approach ensures reliable real-time sensing, remote accessibility, and resilience in electrically noisy agricultural environments.

## 3.2 Architectural Design

### 3.2.1 High level Architectural Design

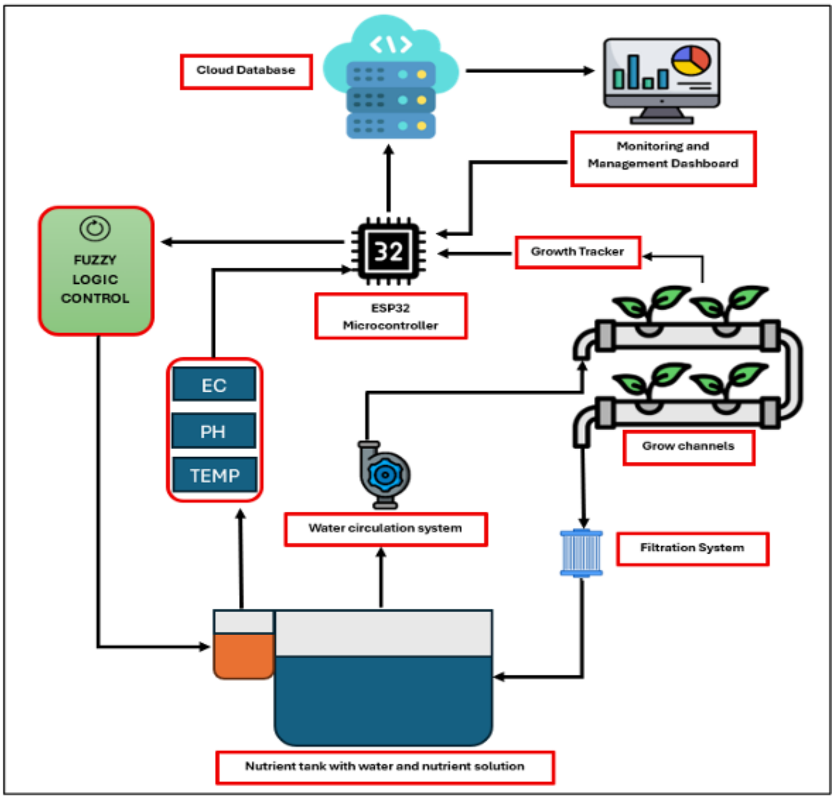


*Figure 1- High level Architectural Design*

### 3.2.2 Hardware and software requirements with justification

**Hardware Requirements**

- **ESP32 Microcontroller**: Selected as the central controller due to its built-in Wi-Fi capability, sufficient processing power for real-time control, low power consumption, and compatibility with IoT applications.
- **RS485 Multi-Parameter Sensor (EC, pH, NPK)**: Chosen for its industrial reliability and noise-resistant communication, ensuring accurate nutrient measurements in electrically noisy environments.
- **Ultrasonic Sensor**: Used for non-contact plant height measurement, avoiding physical damage to plants while enabling growth trend monitoring.
- **Water Level Sensor**: Essential for enforcing safety constraints and preventing pump dry-run conditions.
- **Relay Modules**: Provide electrical isolation between the low-voltage microcontroller and high-voltage pumps.
- **Nutrient and pH Dosing Pumps**: Enable controlled and precise chemical dosing required for inorganic NFT hydroponics.

**Software Requirements**

- **Embedded Firmware (C/C++ using Arduino IDE)**: Enables efficient development, testing, and maintenance of control logic.
- **Firebase Realtime Database**: Used for real-time cloud data storage, synchronization, and remote accessibility.
- **Mobile Application Dashboard**: Provides a user-friendly interface for monitoring sensor data and controlling pumps remotely.

These hardware and software components were selected to balance performance, reliability, cost, and ease of integration.

### 3.2.3 Risk Mitigation Plan with alternative solution identification

Several potential risks were identified during the system design phase, along with appropriate mitigation strategies:

- **Sensor Failure or Drift**: Inaccurate sensor readings may lead to incorrect control actions. This risk is mitigated by periodic calibration and cross-verification of sensor data. As an alternative, redundant sensors or manual override controls can be introduced.
- **Low Water Level Conditions**: Insufficient nutrient solution may cause pump damage or incorrect dosing. This risk is mitigated through a hard safety cutoff that disables all pumps when low water levels are detected. An alternative solution is the integration of automatic refill mechanisms.
- **Network Connectivity Loss**: Loss of Wi-Fi connectivity may prevent cloud synchronization. The system mitigates this risk by maintaining full local autonomous operation independent of cloud availability. An alternative is the use of local data buffering or GSM-based connectivity.
- **Overdosing of Nutrients or pH Solution**: Excessive dosing may harm plant roots. This risk is mitigated using short, time-buffered dosing cycles followed by re-measurement. As an alternative, advanced control algorithms such as fuzzy logic can be implemented in future versions.

### 3.2.4 Cost Benefit Analysis for the proposed solution

The proposed system is designed as a low-cost automation solution using commercially available components while delivering significant operational benefits. The total hardware cost remains affordable for small-scale farmers and academic research setups compared to commercial hydroponic controllers.

From a benefit perspective, the system reduces manual labor, minimizes nutrient wastage, and improves consistency in nutrient management. Automated control decreases the likelihood of human error and enables continuous operation without constant supervision. The use of cloud-based monitoring further enhances convenience and system transparency.

Overall, the benefits of improved nutrient stability, enhanced safety, and reduced operational effort outweigh the initial implementation cost, making the system a cost-effective and practical solution for inorganic NFT hydroponic applications.

### 3.3 Performance requirements

- Sensor data acquisition shall occur at **regular intervals of 5–10 seconds** for critical parameters such as EC, pH, water level, and temperature.
- When corrective action is required, **nutrient or pH dosing pumps shall be activated within 1–2 seconds** of control logic execution.
- After each dosing operation, a **minimum stabilization and mixing delay of approximately 2 minutes** shall be enforced before re-evaluating sensor readings.
- Sensor readings and system status shall be uploaded to the cloud in **near real time**, with an acceptable delay of a few seconds depending on network conditions.
- The embedded firmware shall operate within the **limited RAM and flash memory of the ESP32**, using lightweight libraries and efficient data handling.
- **No long-term data storage** shall be maintained locally; historical data storage and analysis shall be handled by the **cloud platform** to minimize runtime memory usage.

## 3.4 Design constraints

- The system is implemented as a **prototype-scale embedded solution**, limiting large-scale deployment and mechanical robustness.
- Design choices are constrained by the **processing power, memory capacity, and I/O availability** of the ESP32 microcontroller.
- Sensor accuracy and response times are limited by the **specifications of commercially available EC, pH, NPK, and ultrasonic sensors**.
- The system relies on **pre-mixed inorganic nutrient solutions (A+B)**, restricting fine-grained individual nutrient control.
- Cloud-based features depend on **network connectivity**; therefore, the system prioritizes **local autonomous operation** during connectivity loss.
- Advanced mechanical design features, such as industrial-grade enclosures, are **outside the scope** of the current implementation.

## 3.5 Software system attributes

### 3.5.1 Reliability

The system is designed to operate reliably under continuous conditions with minimal human intervention. Reliability is achieved through rule-based control logic, deterministic execution on the ESP32 microcontroller, and the use of time-buffered dosing actions to prevent oscillation or shock dosing. Safety-critical checks, such as water-level validation, are executed with the highest priority to ensure correct operation even when sensor anomalies occur.

Functions most likely to change in future versions include sensor calibration routines and control threshold values, as these may be refined based on long-term operational data or different crop requirements.

### 3.5.2 Availability

The system is intended to operate continuously on a 24/7 basis during the plant growth cycle. Core nutrient monitoring and control functions are designed to remain fully operational even in the absence of cloud connectivity. Local autonomous operation ensures availability of essential functions such as EC and pH regulation regardless of network conditions.

Cloud-dependent features, such as remote visualization and manual override via the mobile dashboard, may experience reduced availability during network outages but do not affect core system functionality.

### 3.5.3 Security

- Access to the cloud database is restricted through authenticated Firebase credentials to prevent unauthorized reading or modification of system data.
- Communication between the embedded system and the cloud platform is limited to predefined database paths and validated requests only.
- All control commands received from the remote dashboard are validated by the embedded firmware before execution to avoid unintended or unsafe pump activation.
- Relay modules are used to electrically isolate the low-voltage ESP32 control circuitry from high-voltage pump power lines, ensuring electrical safety and protecting the controller from faults.
- Safety-critical conditions, such as low water level detection, override all remote and local control commands, preventing pump operation under unsafe conditions.

### 3.5.4 Maintainability

The embedded software is structured in a modular manner to support easy maintenance and future updates. Separate functional modules are used for sensor acquisition, control logic, safety enforcement, actuation, display handling, and cloud communication. This modular structure allows individual components to be modified or extended without impacting the entire system.

Functions expected to change most frequently include sensor calibration logic, control thresholds for different growth stages, cloud data schema definitions, and dashboard interaction logic. These components are intentionally isolated to simplify updates, testing, and long-term system evolution.

## 3.6 Other requirements

- The system shall support **configurable EC and pH threshold values** to accommodate different plant growth stages without requiring code-level modification.
- The system shall provide a **safe manual override mechanism** through the remote dashboard for maintenance and emergency situations.
- The system shall tolerate **short-term power interruptions** by safely stopping all actuation and resuming normal monitoring upon restart.
- The system shall prevent **data corruption or unsafe actuation** during power recovery.
- The system shall generate **alerts and notifications** for abnormal operating conditions to support timely human intervention.

# 4. Supporting information

## 4.1 Appendices

Appendix A: Sample User Interface Screens shot

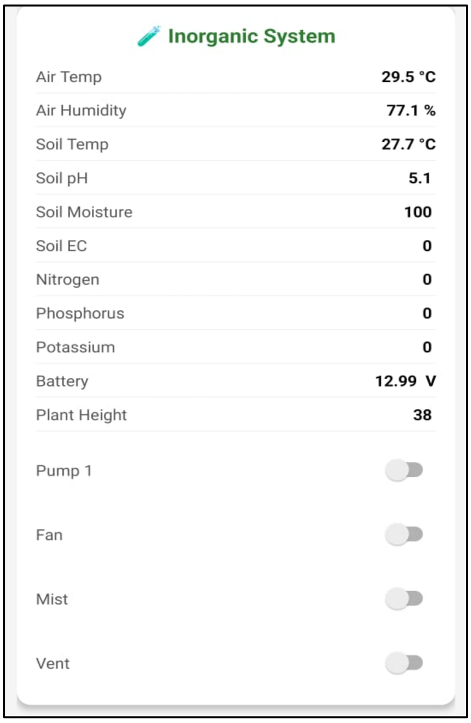


*Figure 2 - Mobile dashboard displaying inorganic system sensor readings*

Appendix B: Inorganic NFT hydroponic system

*Figure 3- Inorganic NFT System*

Appendix C – Hardware Components

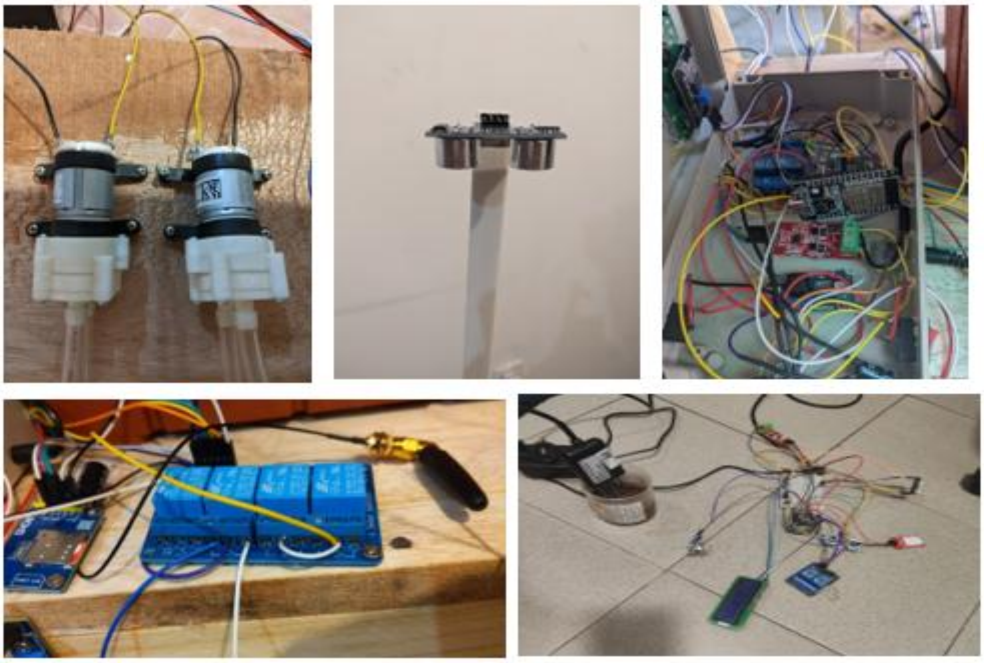

*Figure 4 - Hardware Components*

# 5.References


[1] D. Resh, *Hydroponic Food Production: A Definitive Guidebook for the Advanced Home Gardener and the Commercial Hydroponic Grower*, 7th ed. Boca Raton, FL, USA: CRC Press, 2013.

[2] Espressif Systems, *ESP32 Technical Reference Manual*, Espressif Systems Inc., 2023. [Online]. Available: https://www.espressif.com

[3] Google LLC, *Firebase Realtime Database Documentation*, 2024. [Online]. Available: https://firebase.google.com/docs/database

[4] R. Jensen and J. Collins, "Nutrient management strategies in NFT hydroponic systems," *Journal of Controlled Environment Agriculture*, vol. 12, no. 3, pp. 45–53, 2020.